

Sagar Patel

Forward Deployed Engineer

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#) | [Twitter](#)

SKILLS

Programming Languages: C, C++, Python, Bash

Embedded Systems: Embedded Linux, RTOS, Device Driver Development, BSP Integration, Interrupt Handling, Memory Management, Bootloader Development, Firmware Debugging & Optimization, Yocto, Buildroot, Cross-compilation Toolchains, IPC Mechanisms, I2C, SPI, UART, CMake, BIOS

Edge AI & NVIDIA: CUDA, TensorRT, cuDNN, ONNX, NVIDIA Jetson (Xavier, Orin), JetPack SDK, DeepStream, DOCA, NVIDIA GR00T, Nsight Systems

Networking & Data Plane: TCP/IP, Ethernet Switching & Routing, Layer 2/3 Networking, Packet Processing Pipelines, DMA Optimization, Network Telemetry, DPDK, P4

Tools & Infrastructure: Git, GitHub, Docker, CI/CD, Claude Code, Codex, Vercel, VS Code, Cursor, Eclipse IDE

Web & Full Stack: Next.js, React, FastAPI, Supabase, PostgreSQL, Tailwind CSS

Hardware: NVIDIA Jetson Platforms, NVIDIA BlueField DPUs, Raspberry Pi, ESP32, BeagleBone Black, STM32, ARM-based Systems

WORK EXPERIENCE

Ciena

May 2024 – Present

Senior Software Engineer

Remote, USA

- Built and maintained embedded Linux networking firmware on NVIDIA Jetson Orin, optimizing packet processing pipelines and DMA paths to improve sustained throughput by 31%.
- Developed edge AI inference pipelines using CUDA and TensorRT for real-time anomaly detection, reducing unplanned operational downtime by 22%.
- Owned full platform lifecycle — BSP upgrades, secure boot, CI validation, and release management — cutting hardware bring-up timelines by 24%.

Cisco Systems

Jan 2023 – Apr 2024

Software Engineer

Austin, TX

- Engineered P4-programmable switching firmware integrated with ASIC SDKs, improving forwarding stability and reducing packet loss by 31% at scale.
- Implemented AI-assisted traffic intelligence on NVIDIA BlueField DPUs using DOCA and ONNX, improving anomaly classification accuracy by 24% and reducing root-cause analysis time by 34%.
- Developed embedded Linux platform components including secure boot, telemetry drivers, and CI pipelines across multi-router carrier platforms.

Tata Consultancy Services (TCS)

Jul 2020 – Aug 2021

Embedded Software Engineer

Remote, India

- Developed bare-metal RTOS firmware for ARM-based telecom edge nodes, implementing QoS traffic shaping and multicast packet replication to boost delivery rates by 28%.
- Wrote custom kernel modules and device drivers for high-speed Ethernet interfaces, cutting packet jitter by 35% in distributed network deployments.

PROJECTS

Watchpoint

[GitHub](#)

Python · Go · ROS2 · FastAPI · Next.js · Docker

- Built an end-to-end incident analysis platform for ROS2 and edge AI robots — capturing telemetry across logs, metrics, ROS2 topics, inference timing, and hardware state into replayable failure bundles.
- Developed a lightweight Go edge agent cross-compiled for Linux/ARM (Jetson, Raspberry Pi) and a ROS2 Python collector for real-time topic rate and node health monitoring.
- Implemented AI-assisted root-cause analysis covering resource contention, thermal throttling, version regressions, and failure chains with downloadable replay bundles for field diagnostics.

FieldFix

[Portfolio](#)

Edge AI · Python · Embedded Linux · FastAPI

- Built an AI-powered field service application with zero cloud dependencies — designed to run fully offline on edge hardware in environments without reliable network connectivity.
- Architected on-device inference pipeline for real-time diagnostics and decision support in industrial field conditions.

HydraSwarm **Hackathon Winner**

[GitHub](#)

TypeScript · Next.js · OpenAI GPT-4o · HydraDB

- Built a 7-agent AI system where specialized agents (PM, Architect, Developer, Reviewer, QA, SRE, CTO) collaborate on engineering tasks with shared institutional memory via HydraDB.
- Implemented a self-improving pipeline where each run stores lessons back to memory, producing measurably better outputs across subsequent runs (score improves 7→8→9 across runs).
- Shipped with SSE streaming for live agent updates, a full memory explorer UI, run comparison view, and 326 unit tests across 21 suites.

SignalForge

[GitHub](#)

TypeScript · Python · FastAPI · Next.js

- Built a personal intelligence engine for engineers and founders in Edge AI, Robotics, Physical AI, and Embedded Systems — surfacing curated domain-specific technical signal.
- Full-stack application with a Python/FastAPI backend and TypeScript/Next.js frontend deployed on Vercel.

HACKATHONS & AWARDS

Intelligence at the Frontier Hackathon

Jan 2026

- 1st Place — Best Overall Use of DeepLake, Physical AI & Robotics: Data at Scale (XG1)
- 1st Place — New Project Winner, Physical AI & Robotics by NomadicML (XG1)
- 1st Place — Hackathon Winner (HydraSwarm)

Best Capstone Project Award

Dec 2022

University of California, Irvine

- Autonomous Drone for GPS-less Inspection — built on Tarot 650 frame with NVIDIA Jetson Nano, ROS Noetic, and Intel RealSense T265; implemented SLAM, path planning, and obstacle avoidance for indoor/outdoor environments.

CERTIFICATIONS

NVIDIA-Certified Associate: Generative AI LLMs (NCA-GENL) — NVIDIA, 2026

EDUCATION

Sofia University

Jan 2026 – Jun 2027

M.S. in Computer Science

CA, USA

University of California, Irvine

Aug 2021 – Dec 2022

M.S. in Embedded and Cyber Physical Systems

CA, USA

Charotar University of Science and Technology

Jul 2017 – May 2021

B.Tech, Electronics and Communication Engineering

Gujarat, India